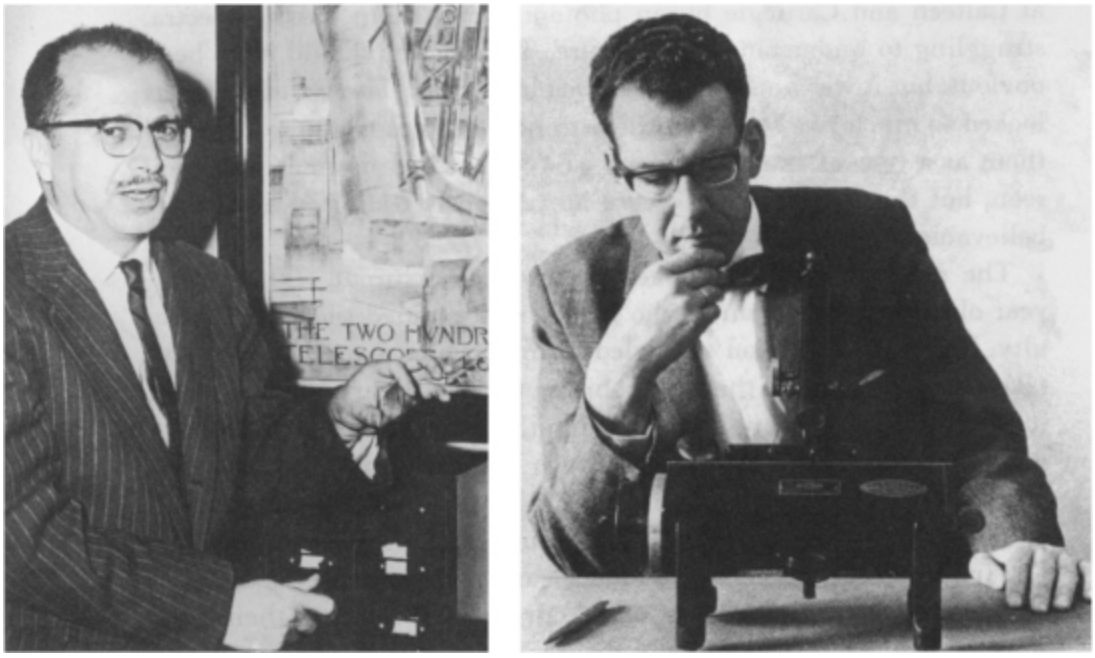


Left: **Jesse L. Greenstein with a drawing of the Palomar 5-meter optical telescope, ca. 1955.** ***Right:*** **Maarten Schmidt, with an instrument for measuring spectra made by the 5-meter telescope, ca. 1963.** [Courtesy the Archives, California Institute of Technology.]

Recall that the expansion of the Universe is like the expansion of the surface of a balloon that is being blown up. If a number of ants are standing on the balloon's surface, each ant will see all the other ants move away from him as a result of the balloon's expansion. The farther away another ant is, the faster the first ant will see it move. Similarly, the farther away a distant object is from Earth, the faster we on Earth will see it move as a result of the Universe's expansion. In other words, the object's speed is proportional to its distance. Therefore, from the speeds of 3C273 and 3C48, Schmidt and Greenstein could infer their distances: 2 billion light-years and 4.5 billion light-years, respectively.

These were enormous distances, nearly the largest distances ever yet recorded. This meant that, in order for 3C273 and 3C48 to be as bright as they appear in the 5-meter telescope, they had to radiate enormous amounts of power: 100 times more power than the most luminous galaxies ever seen.

3C273, in fact, was so bright that, along with many other objects near it on the sky, it had been photographed more than 2000 times since 1895 using modest-sized telescopes. Upon learning of Schmidt's discovery, Harlan Smith of the University of Texas organized a close examination of this treasure trove of photographs, archived largely at Harvard, and discovered that 3C273 had been fluctuating in brightness during the past